FlexControl Knob for Thetis

Version 1.0.0 · by KW4EX · Read Me First

Written for radio operators, not programmers. Every step tells you exactly what to click. The installer does the technical parts for you.

No warranty — use at your own risk. This program is shared free by one radio operator, **as is**. There is **no warranty of any kind**: no promise that it works, that it is safe, or that it suits your equipment or your purpose. Amplifiers, radios and antennas can be damaged, and RF can be dangerous. TUNE and MOX on the knob really transmit. You are responsible for checking that it behaves correctly with your own station, and for operating within your license. The author is not liable for any damage or loss. It is not made, approved or supported by FlexRadio, Thetis, Thetis HL2 or SDR-VST3. Licensed under the MIT License (see LICENSE.txt).

1. What it is

A small program for **Windows 10 and 11** that lets the **FlexRadio FlexControl** USB tuning knob control **Thetis**, **Thetis HL2** and **SDR-VST3**. It needs **no amplifier** and no other radio hardware: just the knob and Thetis.

It is the knob part of the KPA1500 Thetis Bridge (v2.31) as a stand-alone program, for operators who have the knob but not the Elecraft KPA1500 amplifier. It adds a small web page that shows what turning the knob does right now, and what each press does. It can also add a small window inside Thetis that shows the same thing.

2. What it does

- **Turning** the knob tunes VFO A in Thetis' own tuning step. A press can switch the knob to VFO B, AF volume, AGC gain, RIT, XIT, tuning step or TX drive. The three lights on the knob show which.
- **12 presses** (the knob press and buttons 1–3, each as a tap, double tap or long press). Each runs a function you choose on the panel.
- **A panel in your browser** (http://127.0.0.1:8090) shows what the knob does right now and lets you choose the functions.
- **An optional window inside Thetis** ("FlexControl Knob") shows the same thing.

Press	Tap	Double tap	Long press
Knob press	AF volume ⇄ VFO A	Mute	Back to VFO A
Button 1	AF volume ⇄ VFO A	AGC gain ⇄ VFO A	Mute
Button 2	VOX	Tuning step larger	Tuning step smaller
Button 3	TUNE (transmits!)	Band up	Band down

These are the starting settings (the same as the bridge's, except that button 3 tap is a plain TUNE here). Change any of them on the panel.

3. What you need

Item	Details
Windows	Windows 10 or 11.
Thetis	Thetis 2.10.3.12 or newer, Thetis HL2 2.10.3.15 or newer, or SDR-VST3 5.x, installed and working.
Thetis TCI turned on	A setting inside Thetis (port 50001). The install steps below show where.
FlexControl knob	Plugged into this computer with its USB cable.
Python 3.8 or newer	Free. The installer checks for it and offers to install it.
pyserial	A small free helper that reads the knob. The installer installs it (needs internet).

Only one program can use the knob at a time. Close DDUtil, and switch off the knob feature of the KPA1500 Thetis Bridge if you have it (Part 6), before you use this program.

4. What's in this download

File	What it is
READ ME FIRST ...pdf	This guide.
FlexKnob_for_Thetis_v1.0.0.zip	The program. Right-click it, choose Extract All..., then Extract.
LICENSE.txt	The license (MIT).

Inside the zip is a folder **FlexKnob_for_Thetis**. In it:

File	What it is
1_RUN_THIS_Installer.bat	Double-click this. It does the whole install and asks a few easy questions.
README_Install.txt	The install steps as a plain text file.
FlexKnob_Help.html	The full help (open it in your browser).
manual\	The program's own files, for anyone who wants to install by hand.

5. Install, step by step

Part A — Get ready (5 minutes)

1. **Close any other program that uses the knob.** That means DDUtil (if you use it with the knob) and the knob feature of the KPA1500 Thetis Bridge. Part 6 below shows how to switch the bridge's knob off. The installer can do that for you.
2. **Plug the knob into the computer** with its USB cable. Windows normally sets it up by itself (it shows up in Device Manager under Ports).
3. **Turn on Thetis' TCI server.** Start Thetis. Click **Setup** › **Serial/Network/Midi CAT** › the **Network** tab. Set the TCI Server to **127.0.0.1:50001**, tick **TCI Server Running**, click **Apply**, then **OK**.
4. **Close Thetis the normal way** (File › Exit). It is happy to be closed now; the installer may add a window to it.

Part B — Run the installer

1. In your Downloads folder, right-click **FlexKnob_for_Thetis_v1.0.0.zip** › **Extract All...** › **Extract**.
2. Open the new **FlexKnob_for_Thetis** folder and double-click **1_RUN_THIS_Installer.bat**.
3. If Windows says **"Windows protected your PC"**, click **More info**, then **Run anyway**. (The installer is a plain text file; you can open it in Notepad to read exactly what it does.)
4. A window with text opens. It asks a few questions. **Press Enter to accept the answer shown in [brackets]** unless this guide says otherwise.
5. **Step 1 of 6 — Python.** If it says "Found Python...", nothing to do. If not, answer **y** to let it install Python, or install it yourself from python.org (tick **Add python.exe to PATH**) and double-click the installer again. It then installs pyserial by itself (internet needed).
6. **Step 2 of 6 — Install folder.** Press **Enter**. The program goes into **C:\RadioScripts\FlexKnob_for_Thetis**, its own folder, next to your other programs. (Do not choose "Program Files".)
7. **Step 3 of 6 — Only one program can use the knob.** If you have the KPA1500 Thetis Bridge, the installer finds it and offers to switch off its knob. Press **Enter** (yes). Nothing else in the bridge changes. If the bridge is running, the installer shows you the steps to do it yourself (also in Part 6).
8. **Step 4 of 6 — Thetis, start file, shortcuts.** "Should the start file also launch... Thetis?" — press **Enter** (no) to start Thetis your usual way, or **y** to have one shortcut start both. "Put a shortcut on the Desktop?" — **Enter** (yes). "Start automatically when you sign in to Windows?" — **Enter** (no).
9. **Step 5 of 6 — Self-test.** Press **Enter**. After about 30 seconds you should see **ALL TESTS PASSED**. This test uses a pretend knob and a pretend Thetis; nothing real is touched.
10. **Step 6 of 6 — The knob window inside Thetis (optional).** Press **Enter** (yes) to add it. If Thetis is running the installer offers to close it (**Enter**; make sure the radio is not transmitting). A backup of Thetis' settings is saved first. Choose **n** to skip; you can add it later.
11. At the end it offers to **start the knob program now** and opens the panel in your browser. Press **Enter**.

Part C — Check that it works

1. A window titled **FlexControl Knob for Thetis** is open (the knob program). **Leave it open** while you use the radio. To stop the program, click that window and press **Q**.
2. Start **Thetis** (or Thetis HL2 / SDR-VST3) the way you always do. Its TCI server must be on (Part A, step 3).

3. Open the panel in your browser: **http://127.0.0.1:8090** (or double-click **Open_Knob_Panel.bat** in the install folder). At the top you should see two green labels: **Knob connected on COM…** and **Thetis connected (TCI)**.
4. **Turn the knob.** VFO A should tune. The big text on the panel says **VFO A tuning**.
5. **Tap button 1.** The first light on the knob comes on, the panel says **AF volume (master)**, and turning now changes the volume. Tap button 1 again to go back to VFO A.
6. If you added the window to Thetis, it shows the same information. If it says **FLEXKNOB PROGRAM NOT RUNNING**, the knob program is not running.

Part D — Choose what each press does

1. On the panel, find **What each press does**. There is one line for each of the 12 presses.
2. Click the drop-down on a line and pick a function. It is saved at once.
3. Click **Try** to run that function as if you had pressed the knob. (Functions that transmit ask you first.)
4. To go back to the starting settings, click **Put all functions back to the defaults**.

Every day after that: double-click the **FlexKnob for Thetis** shortcut on your Desktop (or **start_flexknob.bat** in C:\RadioScripts\FlexKnob_for_Thetis), then start Thetis.

6. If you also have the KPA1500 Thetis Bridge

The bridge has its own knob feature. Only **one** program can open the knob, so switch the bridge's knob **off** and use this program for the knob. The bridge's amplifier features are not affected.

Switch the knob off in the bridge

1. Start the KPA1500 Thetis Bridge. Open your browser and go to **http://127.0.0.1:8080** (the bridge panel).
2. Click **Settings**, then find **FlexControl tuning knob**.
3. Set the **Knob** drop-down to **Off (not used)**. It saves by itself and works at once. You do not need to restart the bridge.
4. Optional: also untick **Show the KPA1500 Knob window in Thetis** (Settings › Thetis windows › Knob window) and press **Rebuild Thetis windows**, so Thetis does not show two knob windows.
5. Now start the knob program from this download. The panel's first label should turn green: **Knob connected on COM…**.

To go back to the bridge's knob

1. Close the knob program here (click its window and press **Q**).
2. In the bridge panel set **Knob** back to **Find it automatically**.

The two programs use different names and ports, so they do not overwrite each other: this program's panel is at port 8090 (the bridge's is 8080), its Thetis window is called "FlexControl Knob for Thetis - KW4EX" (the bridge's is "KPA1500 Knob - KW4EX"), and each has its own settings file. One thing to know: each program's **Undo** puts back the Thetis settings saved before ITS last change, so using one program's Undo can also take away the other's window. Just run that program's Setup again.

7. Knob modes and lights

What turning the knob does is called its **mode**. The three small lights on the knob (buttons 1-2-3) show the mode:

Turning the knob changes	Lights (1 · 2 · 3)
VFO A tuning	○ ○ ○
VFO B tuning	● ○ ●
AF volume (master)	● ○ ○
AGC gain (RX1)	○ ● ○
RIT (10 Hz per click)	○ ○ ●
XIT (10 Hz per click)	○ ● ●
Tuning step size	● ● ○
TX drive	● ● ●

● = on, ○ = off. VFO, RIT, XIT and the tuning step use Thetis' own step. Spun fast, the VFO moves faster (**Fast spin** on the panel).

8. Safety rules built in

- **TUNE** and **MOX** really make Thetis transmit. Check your antenna or dummy load first. This program does **not** check for an antenna.
- **TUNE** is refused if the radio is already transmitting. It **stops by itself after 20 seconds** (you can change this on the panel; 0 = never). Pressing it again stops it sooner. The panel shows a big **STOP TUNE** button while it is on.
- If Thetis is closed or the link is lost, knob presses that need Thetis do nothing and the panel says so.
- The panel can only be opened from this computer (127.0.0.1), and only this computer's own web address is accepted.
- **Only one copy** of the program runs at a time; a second one says so and stops.

9. Updating and removing

Updating to a newer version

1. Download the new zip and extract it. Each version has its own name with the version number.
2. Close the knob program (press **Q** in its window). The installer can also do this for you.
3. Double-click the new **1_RUN_THIS_Installer.bat**. Choose **Continue**, and choose **Keep your current start options**. Your settings (what each press does) are kept.

Removing it

1. Double-click **Uninstall_FlexKnob_for_Thetis.bat** in C:\RadioScripts\FlexKnob_for_Thetis.
2. Answer **y** to remove the knob window from Thetis (Thetis must be closed; a backup is saved first), and **y** to delete the program's own files. Nothing else in the folder is touched.

10. If something goes wrong

What you see	What to do
"Windows protected your PC"	Click **More info**, then **Run anyway**.
The installer says Python was not found	Install Python from python.org, tick **Add python.exe to PATH**, then double-click the installer again.
Panel: "pyserial is not installed"	Open Command Prompt and type `py -m pip install pyserial` (internet needed), then start the program again.
Panel: "Knob: not found (is it plugged in?)"	Plug the knob in. Double-click **Find_My_Knob.bat**: it says whether Windows sees the knob and on which COM port. If it is found, pick that port in the panel's **Knob settings**.
Panel: "... is in use by another program"	Another program has the knob. Close DDUtil, and switch off the knob in the KPA1500 Thetis Bridge (Part 6). Wait a few seconds.
Panel: "Thetis: waiting..."	Start Thetis, and check Setup › Serial/Network/Midi CAT › Network: **TCI Server Running** ticked, 127.0.0.1:50001, Apply, OK.
"...needs a newer Thetis for the knob functions"	Your Thetis is too old to take these commands over TCI. Update Thetis (2.10.3.12 or newer; HL2 2.10.3.15 or newer). The lights and the mode switching still work.
The panel does not open	The knob program is not running. Start it (Desktop shortcut). If its window says port 8090 is busy, edit start_flexknob.bat and add `--web-port 8091` to the KNOB_FLAGS line.

What you see	What to do
Window in Thetis says "FLEXKNOB PROGRAM NOT RUNNING"	Start the knob program. The window turns normal within a second or two.
Window in Thetis is missing	Close Thetis, double-click **Setup_Thetis_Window.bat**. If Thetis does not show it, in Thetis open Setup › Appearance › Meters/Gadgets and look for "FlexControl Knob for Thetis - KW4EX".
Something is wrong after the Thetis window setup	Close Thetis, double-click **Undo_Thetis_Window.bat**. It puts back the settings saved just before.
The self-test did not pass	Open **install_selftest.txt** in the install folder and send it to whoever gave you the program.

The program's log is **flexknob.log** in the install folder (it keeps itself small). For more detail, add --verbose to KNOB_FLAGS in start_flexknob.bat.

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